

COMP5450 - Exercise 3: Messages Directory App

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WHAT THIS APP DOES

This is a React Native app that shows a directory of message categories. When you open the app, you see 6 colored circles: You, Home, Love, Family, Friends, and School. Each circle has a picture icon inside it on a white background. When you tap a circle, it opens a new screen that shows the messages saved in that category.

I built this for Exercise 3 of COMP5450. The app uses TypeScript and React Navigation to move between screens. It is responsive and works on phones, tablets, and web browsers.

HOW TO RUN THIS IN INTELLIJ IDEA

Step 1: Install What You Need

Before you start, make sure you have these installed on your computer:

1. Node.js 18 - Download from <https://nodejs.org/dist/v18.20.4/node-v18.20.4-x64.msi> (Use version 18, not 20)
2. IntelliJ IDEA - Download from <https://www.jetbrains.com/idea/>
3. Expo CLI - Open a terminal and run: `npm install -g expo-cli`
4. Android Studio - Download from <https://developer.android.com/studio> (for the Android emulator)

Step 2: Open the Project in IntelliJ

1. Open IntelliJ IDEA
2. Click File -> Open
3. Find the folder called MessagesDirectory and select it
4. Click OK
5. Wait for IntelliJ to load the project

Step 3: Install the Dependencies

1. In IntelliJ, open the Terminal tab at the bottom
2. Type this command and press Enter:
npm install

```
npx expo install react-native-web@~0.19.6 react-dom@18.2.0 @expo/webpack-config@^19.0.0
```

3. Wait for it to finish. This downloads all the packages the app needs.

4: Start the App

You can run the app in 3 different ways:

Option A: Run in Chrome (Easiest)

1. In the terminal, type:
npx expo start --web
2. Wait for it to start
3. It will open a new tab in Chrome automatically
4. You can see the app running in your browser

Option B: Run on Android Emulator

1. First, open Android Studio and start an emulator:
 - Open Android Studio
 - Click More Actions -> Virtual Device Manager
 - Click the Play button next to a device to start it
2. In IntelliJ terminal, type:
npx expo start

3. Press a on your keyboard when it asks
4. The app will open on the Android emulator

Option C: Run on Your Real Phone

1. Install the Expo Go app from the App Store or Google Play
2. In IntelliJ terminal, type:
`npx expo start`
3. A QR code will show up in the terminal
4. Open the Expo Go app on your phone
5. Scan the QR code
6. The app will load on your phone

PROJECT STRUCTURE

Here is what each folder and file does:

MessagesDirectory/

App.tsx - Main file, sets up navigation

package.json - List of packages the app needs

tsconfig.json - TypeScript settings

src/

screens/

HomeScreen.tsx - Shows the 6 category circles

MessagesScreen.tsx - Shows messages for one category

components/

CategoryCard.tsx - Circular category button

MessageItem.tsx - Message card component

data/

messagesData.ts - Sample messages with my names

types/

index.ts - TypeScript types

constants/

colors.ts - Colors used in the app

assets/

images/ - Category icons (PNG files)

screenshots/ - Screenshots for README

HOW THE APP WORKS

Home Screen

- Shows a title "Messages Directory" at the top
- Shows 6 colored circles in a 2-column grid
- Each circle has:
 - A white square box in the middle with a picture icon
 - The category name below it
- Colors match the assignment picture:
 - You = Orange-Red (#FF4500)
 - Home = Light Blue (#87CEEB)
 - Love = Steel Blue (#4682B4)
 - Family = Purple (#6A5ACD)
 - Friends = Pink (#FF69B4)
 - School = Blue (#00BFFF)
- The icons are centered inside white boxes that match the circle color theme
- Responsive: Circles resize based on screen width

Messages Screen

- Opens when you tap a category
- Shows a header with the category color

- Has a back arrow to go back
- Shows all messages in that category
- Each message shows:
 - Who sent it
 - When it was sent
 - The message text
 - A colored line on the left
- Responsive: Text size and padding adjust based on screen size

PERSONALIZED NAMES I USED

I used names from my life in the messages:

You: Mine, Self

Home: Mom, Dad, Jameel Brother, Zameer Brother, Sister

Love: My_Love, Friends, Fareed, Sajjad, Naser

Family: Dadi, Aunt Sana, Daddy, Homies, Mom

Friends: Kaleem, Zaid, Zara, Mohammed, Sarah

School: Professor Sabah Mohammed, Classmate, Registrar, Advisor, TA

RESPONSIVE DESIGN

The app uses Dimensions from React Native to check screen size:

- Small phones (width < 360px): Smaller circles, smaller text, less padding
- Normal phones (360px - 768px): Standard size
- Tablets (width > 768px): Bigger circles, larger text, more padding

This means the app looks good on:

- iPhone SE (small screen)

- iPhone 14 / Samsung Galaxy (normal screen)
- iPad / Android tablet (large screen)
- Chrome web browser (any size window)

SCREENSHOTS

Home Screen

[assets/screenshots/home_screen.png]

The main screen with all 6 category circles and centered icons

Messages Screen

[assets/screenshots/messages_screen.png]

Messages inside the School category

Running in Chrome

[assets/screenshots/web_view.png]

The app running in Chrome browser

PROBLEMS I HAD AND HOW I FIXED THEM

1. Navigation was not working - I forgot to wrap the app in `NavigationContainer`. Fixed by adding it in `App.tsx`.
2. Icons were not centered - The icon images were off-center in the circles. I added a white square box inside each circle and centered the image inside it using `justifyContent` and `alignItems`.
3. Screen was too small on web - I used `Dimensions` to make the circles resize based on screen width. I check if the screen is small, normal, or tablet size and change sizes.
4. TypeScript errors - I had to add proper types for navigation props using `StackNavigationProp` and `RouteProp`.

5. Expo error on Windows with Node 20 - I got an error about node:sea. I fixed it by using Expo 49 and Node.js 18 instead of Expo 50 and Node.js 20.
6. Grid not centered on big screens - I added `columnWrapperStyle` and `justifyContent center` to center the 2-column grid.
7. Different theme colors - I made sure each circle uses the exact color from the assignment screenshot. The icon box is white so it matches the background theme of each circle.

WHAT I LEARNED

- How to use React Navigation to move between screens
- How to pass data between screens using route parameters
- How to make a grid layout with `FlatList` and `numColumns`
- How to use TypeScript with React Native
- How to style components with `StyleSheet`
- How to make an app responsive using `Dimensions` API
- How to center images inside circular buttons
- How to run a React Native app in Chrome, emulator, and real phone

NOTES

- I tested this on Chrome web browser, Android emulator, and my phone. It works on all three.
- All messages are fake sample data for the assignment, but I used real names from my life.
- The app was built with Expo 49 because it works better with Node.js 18 on Windows.
- The app is responsive and looks good on small phones, normal phones, tablets, and web browsers.
- The icons are centered inside white boxes that match the circle color theme.

